# Chapter 1

# Data tables in SQL

Most organizations keep their data as tables inside a relational database management system (compared to pandas, CSV, or spreadsheets). Developers talk to those systems using a language called SQL ("Structured Query Language").

Some relational database managers are pricey products you may have heard of before, such as Oracle or Microsoft SQL Server. Some are free, such as PostgreSQL or MySQL. These are server software that client programs talk to over a company's network.[1]

There is a library, called `sqlite`, that lets us create files that hold tables. We can use SQL to create, edit, and browse those tables. sqlite is free, fast, and very easy to install. We will use sqlite instead of a networked database management system.

If you look in your digital resources, you will find a file called `bikes.db`. We created this file using sqlite, and now you will use sqlite to access it.

In the terminal, get to the directory where `bikes.db` lives. To open the sqlite tool on that file:

```
> sqlite3 bikes.db
```

(If your system complains that there is no sqlite3 tool, you need to install sqlite. See this website: https://sqlite.org/)

Please follow along: type each command shown here into the terminal and see what happens.

We mostly run SQL commands in this tool, but there are a few non-SQL commands that all start with a period. To see the tables and their columns, you can run `.schema`:

```
sqlite> .schema
CREATE TABLE bike (bike_id int PRIMARY KEY, brand text, size int,
          purchase_price real, purchase_date date, status text);
```

[1] It should be noted that many notetaking and file storage applications allow you to run SQL-like queries to search your files to meet certain criteria. For example, Obsidian, the Markdown notetaking app, has a plugin called Dataview, which allows you to run searches for notes matching certain *metadata* attribute criteria. See more about this here: https://github.com/blacksmithgu/obsidian-dataview

That is the SQL command that we used to create the `bike` table. You can see all the columns and their types.

You want to see all the rows of data in that table?

```
sqlite> select * from bike;
4997391|GT|57|269.61|2009-05-03|rented
5429447|Cannondale|50|215.91|2002-02-17|broken
5019171|Trek|58|251.17|1985-07-11|rented
3000288|Cannondale|57|211.08|1993-01-05|broken
880965|GT|52|281.75|1995-08-02|available
...
```

You will see 1000 rows of data!

The SQL language is not case-sensitive, so you can also write it like this:

```
sqlite> SELECT * FROM BIKE;
```

Often, you will see SQL with just the SQL keywords in all caps:

```
sqlite> SELECT * FROM bike;
```

The semicolon is not part of SQL, but it tells sqlite that you are done writing a command and that it should be executed.

SQL lets you choose which columns you would like to see. The asterisk (FIXME) used above signifies all columns, and the `bike_id, brand` only gets the bike's id and brand from the dataframe:

```
sqlite> SELECT bike_id, brand FROM bike;
4997391|GT
5429447|Cannondale
5019171|Trek
3000288|Cannondale
...
```

Using WHERE, SQL lets you use conditions to decide which rows you would like to see, and can be combined with the common operators `AND`, `OR`, and `NOT`:

```
sqlite> SELECT * FROM bike WHERE purchase_date > '2009-01-01' AND brand = 'GT';
4997391|GT|57|269.61|2009-05-03|rented
326774|GT|56|165.0|2009-06-27|available
```

```
264933|GT|52|302.43|2009-07-09|available
5931243|GT|55|173.56|2009-11-26|rented
4819848|GT|51|221.71|2009-12-11|rented
9347713|GT|52|232.32|2009-06-13|available
3019205|GT|58|262.94|2009-08-22|available
```

Using DISTINCT, SQL lets you get just one copy of each value:

```
sqlite> SELECT DISTINCT status FROM bike;
rented
broken
available

Busted
Flat tire
good
out
Rented
```

You can also edit these rows. For example, if you wanted every status that is `Busted` to be changed to `broken`, you can use an `UPDATE` statement with a `SET`:

```
sqlite> UPDATE bike SET status='broken' WHERE status='Busted';
sqlite> SELECT DISTINCT status FROM bike;
rented
broken
available
Flat tire
good
out
Rented
```

You can insert new rows:

```
sqlite> INSERT INTO bike (bike_id, brand, size, purchase_price, purchase_date,
↪  status)
  ...> VALUES (1, 'GT', 53, 123.45, '2020-11-13', 'available');
sqlite> SELECT * FROM bike WHERE bike_id = 1;
1|GT|53|123.45|2020-11-13|available
```

Note that the bike_id here must be *unique*.

You can delete rows:

```
sqlite> DELETE FROM bike WHERE bike_id = 1;
sqlite> SELECT * FROM bike WHERE bike_id = 1;
```

To get out of sqlite, type `.exit`.

### *Exercise 1* SQL Query

Execute an SQL query that returns the `bike_id` (no other columns) of every Trek bike that cost more than $300.

*Working Space*

*Answer on Page 7*

## 1.1 Using SQL from Python

The people behind sqlite created a library for Python that lets you execute SQL and fetch the results from inside a python program.

Let's create a simple program that fetches and displays the bike ID and purchase date of every Trek bike that cost more than $300.

Create a file called `report.py`:

```
import sqlite3 as db

con = db.connect('bikes.db')
cur = con.cursor()

cur.execute("SELECT bike_id, purchase_date FROM bike WHERE purchase_price > 330
↪  AND brand='Trek'")
rows = cur.fetchall()

today = datetime.date.today()
for row in rows:
  print(f"Bike {row[0]}, purchased {row[1]}")

con.close()
```

When you execute it, you should see:

```
> python3 report.py
Bike 4128046, purchased 2007-08-06
Bike 7117808, purchased 1995-03-12
Bike 7176903, purchased 1986-07-03
Bike 827899, purchased 2009-03-14
Bike 363983, purchased 1970-08-16
```

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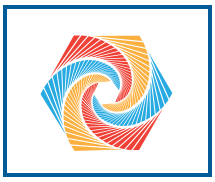
*This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.*

Appendix A

# Answers to Exercises

## Answer to Exercise 1 (on page 4)

```
1 SELECT bike_id FROM bike WHERE purchase_price > 330 AND brand='Trek'
```



# INDEX

SQL, 1
SQLite, 1